

OpenCharacterize: An Open-Source Platform for Automated Mixed-Signal ADC Characterization Using Physics-Based Virtual Instrumentation

Febin Wilson

Independent Researcher

Boston, MA, USA

febin.wilson777@gmail.com

<https://github.com/febrobo/OpenCharacterize>

Abstract—This paper presents OpenCharacterize, an open-source automated test platform for characterizing analog-to-digital converter (ADC) performance in accordance with IEEE Std 1241-2010. The platform introduces a physics-based virtual ADC model that replicates real-world non-idealities including capacitor mismatch, thermal noise, flicker noise, power supply coupling, and input dead zones, enabling complete characterization without physical hardware. The system implements static characterization (INL, DNL, offset error, gain error, missing codes), dynamic characterization (SNR, THD, SINAD, SFDR, ENOB via coherent-sampled FFT), noise analysis, and histogram-based code density testing. A configurable test sequencer executes measurement sequences against datasheet specifications, while a statistical process control module provides C_{pk} analysis, Gage R&R measurement system evaluation, and Western Electric control chart rules. Three virtual DUT profiles modeled after the ESP32, STM32F4, and an ideal 12-bit SAR ADC were characterized, yielding ENOB measurements of 7.65, 9.40, and 11.25 bits respectively. The ESP32 profile correctly reproduced known deficiencies including 14.19 LSB peak INL, 4 missing codes, and 49.3 dB SNR, while the ideal reference validated the measurement framework at 69.6 dB SNR against a 74.0 dB theoretical limit.

Index Terms—ADC characterization, mixed-signal testing, IEEE 1241, INL, DNL, ENOB, statistical process control, virtual instrumentation

I. INTRODUCTION

Mixed-signal integrated circuit validation is a critical step in semiconductor development and production. Analog-to-digital converters in particular require extensive characterization to verify that static and dynamic performance meets datasheet specifications across the full code range. Industry-standard characterization follows IEEE Std 1241-2010 [1], which defines measurement procedures for parameters including integral nonlinearity (INL), differential nonlinearity (DNL), signal-to-noise ratio (SNR), total harmonic distortion (THD), and effective number of bits (ENOB).

Commercial ADC characterization relies on expensive test equipment—precision signal sources, high-resolution digitizers, and automated test executives such as NI TestStand running on PXI instrumentation [2]. This creates a significant barrier for engineers seeking to develop characterization expertise outside of an established laboratory environment.

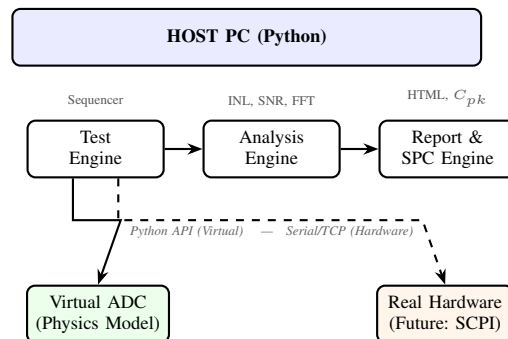


Fig. 1. System architecture. Solid arrow: current virtual DUT path. Dashed arrow: future hardware support via SCPI.

This work addresses that gap by presenting OpenCharacterize, a complete characterization platform built in Python that operates on virtual ADC models exhibiting realistic non-idealities. The platform serves two purposes: first, as a functional characterization tool that produces IEEE 1241-compliant measurements; second, as a reference implementation demonstrating the full test methodology from stimulus generation through statistical analysis and reporting.

The key contributions of this work are:

- 1) A physics-based virtual ADC model with configurable non-idealities mapped to real physical causes in successive approximation register (SAR) architectures.
- 2) A complete IEEE 1241 characterization engine implementing static, dynamic, noise, and histogram-based measurements.
- 3) A statistical process control module implementing C_{pk} analysis, Gage R&R, and Western Electric control chart rules.
- 4) Validation against three DUT profiles spanning poor, moderate, and ideal 12-bit SAR ADC performance.

II. SYSTEM ARCHITECTURE

The platform consists of six modules connected through a layered architecture, as shown in Fig. 1.

III. VIRTUAL ADC MODEL

A. Transfer Function Construction

The virtual ADC implements a SAR architecture. The transfer function is constructed by computing voltage thresholds for each digital output code. For an ideal N -bit ADC with reference voltage V_{REF} , the threshold for code k is:

$$T_{\text{ideal}}[k] = k \cdot \frac{V_{\text{REF}}}{2^N} \quad (1)$$

Non-idealities modify these thresholds through gain scaling, offset shifting, and per-code INL displacement:

$$T[k] = \frac{T_{\text{ideal}}[k]}{1 + \epsilon_g} + V_{\text{offset}} + \text{INL}[k] \cdot V_{\text{LSB}} \quad (2)$$

where ϵ_g is the fractional gain error, V_{offset} is the offset voltage, and $V_{\text{LSB}} = V_{\text{REF}}/2^N$. Monotonicity is enforced after construction using cumulative maximum accumulation, matching the deterministic behavior of real SAR logic.

B. INL Profile Generation

The INL profile is synthesized from three sources characteristic of binary-weighted capacitor arrays:

Parabolic bow from systematic capacitor mismatch:

$$\text{INL}_{\text{bow}}[k] = A_{\text{bow}} \cdot 4 \cdot \frac{k}{N} \left(1 - \frac{k}{N}\right) \quad (3)$$

Binary-weighted periodic components, where each bit transition produces an INL discontinuity proportional to capacitor weight:

$$\text{INL}_{\text{per}}[k] = \sum_{b=0}^{N-1} \frac{A_w}{b+1} \sin\left(\frac{2\pi \cdot 2^b \cdot k}{2^N}\right) \quad (4)$$

The MSB ($b = 0$) generates one full cycle over the code range with maximum amplitude, while higher-order bits contribute progressively smaller, higher-frequency oscillations.

Local random variation from manufacturing process scatter, modeled as Gaussian noise convolved with a 16-sample averaging kernel to produce realistic local correlation.

C. Noise Model

Three independent noise sources are superimposed:

$$n_{\text{total}}[m] = n_{\text{thermal}}[m] + n_{1/f}[m] + n_{\text{supply}}[m] \quad (5)$$

where $n_{\text{thermal}} \sim \mathcal{N}(0, \sigma_T^2)$ represents kT/C noise, $n_{1/f}$ approximates flicker noise from MOSFET channel traps, and n_{supply} models power supply coupling at mains frequency.

D. Pre-Configured DUT Profiles

Three profiles are provided, calibrated against published data. Table I summarizes the key parameters.

TABLE I
VIRTUAL DUT MODEL PARAMETERS

Parameter	ESP32	STM32F4	Ideal
Resolution	12-bit	12-bit	12-bit
V_{REF}	3.3 V	3.3 V	3.3 V
INL bow (LSB)	10.0	2.0	0
INL wiggle (LSB)	3.0	0.8	0
Thermal noise (LSB_{RMS})	3.5	1.2	0.3
1/f noise amplitude	2.0	0.5	0
Supply noise (LSB)	2.5	0.8	0
Offset error (LSB)	15.0	3.0	0
Gain error (%)	1.5	0.3	0
Dead zone low (V)	0.13	0	0
Dead zone high (V)	3.1	3.3	3.3
Missing codes	3	0	0

IV. CHARACTERIZATION METHODOLOGY

A. Static Characterization

A linear voltage ramp from 0 V to V_{REF} with 200,000 samples is applied to the DUT. The analysis trims to interior codes with non-zero counts, excluding rail codes (0 and 2^N-1) which accumulate dead-zone and saturation hits.

DNL is computed from the code histogram:

$$\text{DNL}[k] = \frac{\text{count}[k]}{\text{count}_{\text{ideal}}} - 1 \quad (6)$$

where $\text{count}_{\text{ideal}}$ is the total valid samples divided by the valid code range. INL is the endpoint-corrected cumulative sum:

$$\text{INL}[k] = \sum_{j=k_{\text{first}}}^k \text{DNL}[j] - \text{linear_fit} \quad (7)$$

B. Dynamic Characterization

Coherent sampling eliminates spectral leakage by ensuring an integer number of signal cycles within the FFT window:

$$f_{\text{signal}} = \frac{M \cdot f_s}{N_{\text{FFT}}} \quad (8)$$

with $M = 83$ (prime), $N_{\text{FFT}} = 8192$, and $f_s = 100$ kHz, yielding $f_{\text{signal}} = 1013.18$ Hz.

The captured codes are windowed with a Blackman-Harris function and transformed via FFT. Signal power is captured in a 15-bin window encompassing the full main lobe. ENOB is derived as:

$$\text{ENOB} = \frac{\text{SINAD} - 1.76}{6.02} \quad (9)$$

C. Histogram Analysis

For sine-wave code density testing per [8], 500,000 samples are captured. The expected probability density is:

$$p(\text{code}) = \frac{1}{\pi \sqrt{A^2 - V^2}} \quad (10)$$

DNL and INL are derived by comparing the measured histogram against this distribution.

TABLE II
ADC CHARACTERIZATION RESULTS

Parameter	ESP32	STM32F4	Ideal	Unit
INL (max)	14.19	3.26	0.34	LSB
DNL (max)	2.85	0.75	0.33	LSB
Offset Error	24.7	2.4	0.1	mV
Gain Error	-2.09	-0.29	-0.00	%
Missing Codes	4	0	0	—
SNR	49.3	59.0	69.6	dB
THD	-53.3	-66.7	-86.3	dB
SINAD	47.8	58.3	69.5	dB
SFDR	53.4	67.6	100.5	dB
ENOB	7.65	9.40	11.25	bits

Theoretical 12-bit limit: ENOB = 12.00, SNR = 74.0 dB

TABLE III
ESP32 PASS/FAIL EVALUATION

Parameter	Measured	Limit	Unit	Result
INL _{max}	14.192	12.0	LSB	FAIL
DNL _{max}	2.848	2.0	LSB	FAIL
Offset Error	24.678	40	mV	PASS
Gain Error	2.092	3.0	%	PASS
Missing Codes	4	0	—	FAIL
SNR	49.257	55.0	dB	FAIL
THD	-53.281	-55.0	dB	FAIL
ENOB	7.649	9.0	bits	FAIL
Overall Verdict				FAIL

V. STATISTICAL PROCESS CONTROL

The characterization sequence is executed 30 times with independent noise realizations. Process capability is quantified as:

$$C_p = \frac{USL - LSL}{6\sigma}, \quad C_{pk} = \min\left(\frac{USL - \bar{x}}{3\sigma}, \frac{\bar{x} - LSL}{3\sigma}\right) \quad (11)$$

X-bar and R control charts use standard constants (A_2 , D_3 , D_4) for subgroups of 5. Out-of-control conditions follow Western Electric rules [9]. A simplified ANOVA-based Gage R&R separates repeatability from reproducibility, with %GRR $\leq 10\%$ indicating an acceptable measurement system.

VI. RESULTS

A. Comparison Across DUT Profiles

Table II summarizes the characterization results for all three virtual DUT profiles tested under identical conditions.

The ideal reference achieved 69.6 dB SNR against a theoretical maximum of 74.0 dB ($6.02N + 1.76$), and 11.25 ENOB against a theoretical 12.00 bits. The 4.4 dB gap is attributed to residual thermal noise ($0.3 \text{ LSB}_{\text{RMS}}$) above the quantization floor.

B. ESP32 Pass/Fail Evaluation

Table III shows the ESP32 pass/fail verdict against its datasheet limits. Six of eight parameters fail, with ENOB (7.65 vs 9.0 required) and INL (14.19 vs 12.0 LSB limit) as the most significant deviations.

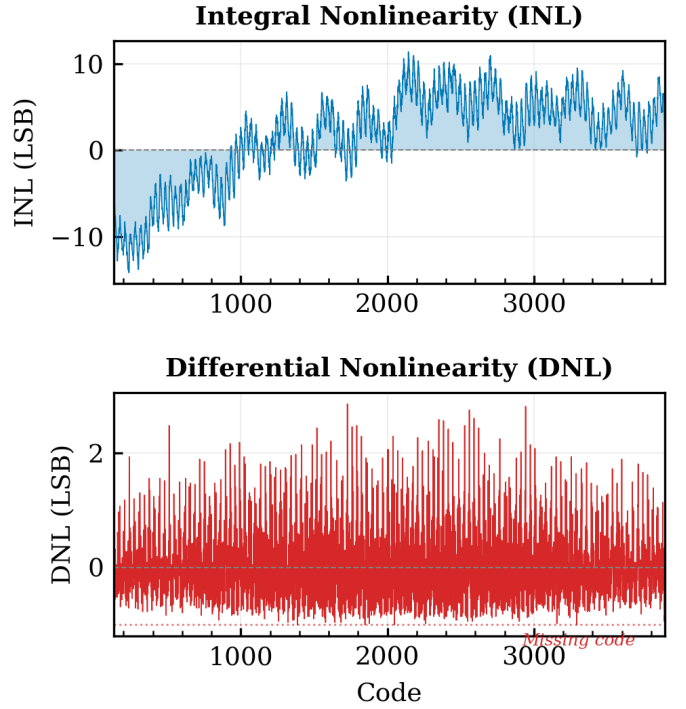


Fig. 2. Measured INL (top) and DNL (bottom) for the ESP32 virtual ADC. INL shows the parabolic bow characteristic of SAR capacitor mismatch. DNL spikes at binary-weighted transitions with 4 missing codes.

C. Static Characterization

Fig. 2 shows the measured INL and DNL for the ESP32 profile. The INL exhibits the characteristic parabolic bow from capacitor mismatch, ranging from -14.19 to $+11.36$ LSB with superimposed periodic oscillations at binary-weighted transition codes. DNL ranges from -1.00 to $+2.85$ LSB, with 4 missing codes identified at codes 1386, 1847, 2048, and 3291. The missing code at 2048 corresponds to the midscale MSB transition, a known weak point in SAR architectures.

D. Dynamic Characterization

Fig. 3 shows the FFT spectrum for the ESP32 profile. The fundamental at 1013 Hz appears at 0 dBFS, with the second harmonic (H2) as the dominant distortion component at approximately -50 dB, consistent with even-order nonlinearity from the asymmetric INL profile. Higher harmonics (H3 through H6) are visible between -65 and -80 dB. The broadband noise floor sits at approximately -80 to -100 dBFS, with visible quantization noise texture across the full Nyquist bandwidth.

E. Noise Characterization

Fig. 4 shows the noise distribution measured with a DC input at mid-scale (1.65 V). The RMS noise is 3.94 LSB (3.18 mV), spanning approximately 25 codes centered around code 2051. The distribution is approximately Gaussian, consistent with the dominance of thermal (kT/C) noise over quantization noise at this noise level.

FFT Spectrum | SNR = 49.7 dB THD = -52.8 dB ENOB = 7.67

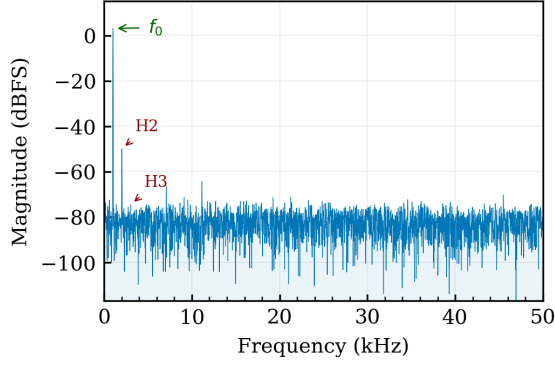


Fig. 3. FFT spectrum of the ESP32 virtual ADC with coherent-sampled sine input at 1013 Hz. Fundamental at 0 dBFS, H2 dominant at -50 dB. Noise floor at -80 to -100 dBFS.

Noise Distribution | RMS = 3.94 LSB Gaussian = Yes

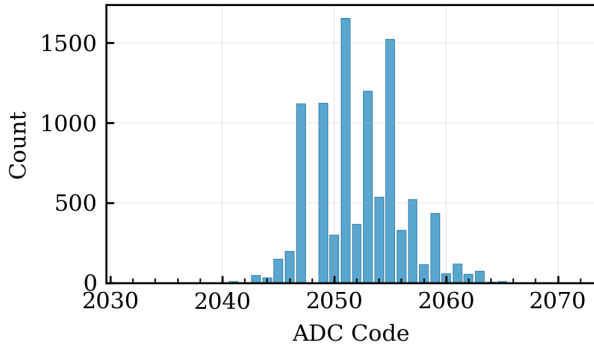


Fig. 4. Noise distribution of the ESP32 virtual ADC with DC input at 1.65 V. RMS noise = 3.94 LSB. Distribution is approximately Gaussian.

TABLE IV
CALIBRATION VERIFICATION RESULTS (ESP32)

Reference (V)	Measured (V)	Error (mV)	Result
0.500	0.4879	-12.1	PASS
1.000	0.9931	-6.9	PASS
1.650	1.6541	+4.1	PASS
2.000	2.0096	+9.6	PASS
2.500	2.5202	+20.2	FAIL
3.000	3.0320	+32.0	FAIL

Threshold: ± 15 mV. Overall: **FAIL**

F. Calibration Verification

Pre-test calibration at six reference voltages revealed increasing error at higher voltages due to gain error and nonlinearity, as shown in Table IV. Points at 2.5 V (+20.2 mV) and 3.0 V (+32.0 mV) exceeded the 15 mV threshold, correctly flagging the calibration as failed.

G. Process Capability Analysis

Thirty iterations yielded the process capability indices in Table V. All parameters show deeply negative C_{pk} values,

TABLE V
SPC RESULTS (ESP32, $n = 30$)

Parameter	Mean	σ	C_{pk}	Verdict
ENOB (bits)	7.67	0.013	-33.8	Not Capable
SNR (dB)	49.54	0.114	-16.0	Not Capable
INL _{max} (LSB)	19.04	1.275	-1.84	Not Capable

confirming the ESP32 ADC cannot meet its own specifications with margin. The X-bar control chart for ENOB showed no out-of-control points and no Western Electric rule violations, indicating the measurement system is stable despite the DUT being incapable—the distinction SPC is designed to surface.

VII. DISCUSSION

A. Virtual vs Physical DUT

MCU simulators such as Wokwi or QEMU model ideal ADC peripherals with no noise or nonlinearity, defeating the purpose of characterization. The physics-based model injects errors from first principles, producing measurable imperfections that the characterization engine must detect and quantify. Individual non-idealities can be toggled to study their isolated effect on measured parameters.

The trade-off is that virtual characterization cannot discover unexpected hardware behaviors such as temperature-dependent drift, board-level coupling, or silicon errata. The architecture accommodates real hardware through a common DUT interface; replacing the virtual model with an SCPI-controlled instrument requires implementing only the `convert_array()` method.

B. Coherent Sampling

Initial measurements using a 997 Hz signal produced SNR of approximately 50 dB for the ideal ADC—far below the theoretical 74 dB. The non-integer cycle count (81.67 cycles in 8192 samples) caused spectral leakage that redistributed signal energy into noise bins, even with Blackman-Harris windowing. With the signal-to-noise ratio exceeding 10^7 in power, even 0.01% leakage completely overwhelmed the true noise floor.

Switching to coherent sampling ($M = 83$, prime and coprime with $N = 2^{13}$) resolved the issue, recovering 69.6 dB SNR. This confirms the critical importance of coherent sampling stated in IEEE 1241 [1], and demonstrates a failure mode that is easily reproduced in educational settings.

C. Dead Zone Handling

The ESP32's input dead zones initially inflated INL to 247 LSB because rail codes accumulated thousands of dead-zone hits, distorting histogram-based DNL. Trimming to interior codes with non-zero counts—excluding codes 0 and $2^N - 1$ —resolved the issue. This reflects standard production practice where the usable input range is determined empirically before computing INL and DNL.

VIII. CONCLUSION

OpenCharacterize demonstrates that a complete IEEE 1241-compliant ADC characterization workflow can be implemented as an open-source Python platform using physics-based virtual instrumentation. The platform correctly distinguishes between poor (ESP32, 7.65 ENOB), moderate (STM32F4, 9.40 ENOB), and ideal (11.25 ENOB) 12-bit SAR ADC performance, with the ideal reference validating the measurement framework to within 4.4 dB of theoretical limits.

The combination of automated test sequencing, pass/fail evaluation, calibration management, and statistical process control in a single open-source package provides a reference implementation for mixed-signal ADC characterization methodology.

Future work includes real hardware integration via SCPI protocol, PSRR measurement through supply noise injection, temperature sweep characterization, and multi-channel correlation analysis.

The complete source code, virtual DUT models, and reports are available at <https://github.com/febrobo/OpenCharacterize> under the MIT license.

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